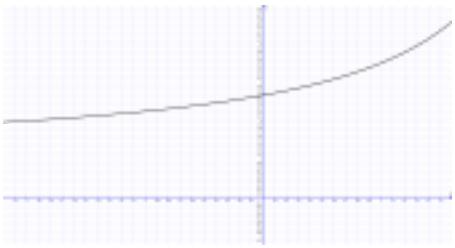
Question	Answer	Marks	Partial Marks
4(a)	462	**1**	
4(b)(i)	$\frac{7}{15}$ oe	**1**	
4(b)(ii)	$\frac{7}{15}\times\frac{6}{14}+\frac{6}{15}\times\frac{5}{14}+\frac{2}{15}\times\frac{1}{14}$ $=\frac{37}{105}$	**3**	**M2** for addition of two of $\frac{7}{15}\times\frac{6}{14}+\frac{6}{15}\times\frac{5}{14}+\frac{2}{15}\times\frac{1}{14}$ or **M1** for one of the products seen
4(b)(iii)	$\frac{29}{65}$ oe	**4**	**M3** for $\frac{7}{15}\times\frac{6}{14}\times\frac{5}{13}+3\times\frac{7}{15}\times\frac{6}{14}\times\frac{6}{13}+3\times\frac{7}{15}\times\frac{6}{14}\times\frac{2}{13}$ oe or $1-3\left(\frac{8}{15}\times\frac{7}{14}\times\frac{7}{13}\right)-\left(\frac{8}{15}\times\frac{7}{14}\times\frac{6}{13}\right)$oe or **M2** for the sum of at least two of $\frac{7}{15}\times\frac{6}{14}\times\frac{5}{13}$, $N\times\frac{7}{15}\times\frac{6}{14}\times\frac{6}{13}$, $N\times\frac{7}{15}\times\frac{6}{14}\times\frac{2}{13}$ seen or for $\frac{7}{15}\times\frac{6}{14}\times\frac{13}{13}$ or $\frac{7}{15}\times\frac{6}{14}+N\times\frac{7}{15}\times\frac{6}{14}\times\frac{k}{13}$ seen or **M1** for $\frac{7}{15}\times\frac{6}{14}\times\frac{5}{13}$ or $N\times\frac{7}{15}\times\frac{6}{14}\times\frac{6}{13}$ or $N\times\frac{7}{15}\times\frac{6}{14}\times\frac{2}{13}$ seen If 0 scored **SC1** for $\frac{1519}{3375}$ oe
5(a)	27[.0] or 26.97… nfww	**3**	**M2** for [cos =] $\frac{8.6^2+9.7^2-4.4^2}{2\times8.6\times9.7}$ or **M1** for implicit form
5(b)	9.19 or 9.192 to 9.193	**4**	**B1** for [angle BCD =] 73 seen **M2** for $\frac{9.7\times\sin 65}{\sin\,(180-65-42)}$ oe or **M1** for $\frac{\sin(180-65-42)}{9.7}=\frac{\sin 65}{DC}$ oe

Question	Answer	Marks	Partial Marks
5(c)	6.15 or 6.149 to 6.151…	**3**	**M2** for $\frac{d}{their\,9.19}$ = sin42 oe or **M1** for right angle between line from *C* to *BD* and *BD* soi
6(a)(i)	[*a* =] 4 [*b* =] – 3 nfww	**2**	**B1** for [*a* =] 4 **B1** for [*b* =] – 3 nfww
6(a)(ii)	*y* = 4 oe	**1**	
6(a)(iii)	*y* = – 6*x* + 7 oe final answer	**2**	**B1** for answer – 6*x* + 7 or answers *y* = – 6*x* + *c* or *y* = *kx* + 7 (*k* < 0)
6(b)(i)	2.25 2.67 3.5	**3**	**B1** for each
6(b)(ii)	correct curve 	**4**	**B3 FT** for 7 or 8 points or **B2 FT** for 5 or 6 points or **B1 FT** for 3 or 4 points
6(c)(i)	–0.78 to –0.72 and 0.55 to 0.59	**2**	**B1** for each
6(c)(ii)	$3x^3 - 9x^2 - 3x + 4$ [= 0] final answer	**4**	**B3FT** for 3 out of 4 correct terms or for $bx^3 - 3bx^2 + (a - 1)x + 8 - 3a$ [= 0] oe or **B2FT** for 2 out of 4 correct terms or for 3 out of 4 terms from $bx^3 - 3bx^2 + (a - 1)x + 8 - 3a$ [= 0] or **M1** for $1 + \frac{5}{3-x} = their\ 4 + (their(-3))x^2$ oe
7(a)(i)	70	**1**	
7(a)(ii)	78	**1**	
7(a)(iii)	Value in range 86 < *V* ≤ 90	**1**	